

Python Programming Exercises (Punch & Enbody Chap 9)

1. Given `D = {'a':3, 'x':7, 'r':5}`:
 - (a) Write Python code that returns the value at key 'x'.
 - (b) Write Python code that returns the key given value '7'.

2. If you had two lists, one of first names and one of last names `['Jane', 'John', 'Jack']` and `['Doe', 'Deer', 'Black']`, use `zip` to create a dictionary with the keys as the first names and the values as the last names.

3. If `my_set = 'bcd'` and `your_set = 'abcde'`:
 - (a) What is the value of `my_set.issubset(your_set)`?
 - (b) What is the value of `your_set.issubset(my_set)`?

4. If `my_dict = {'a':15, 'c':35, 'b':20}`, write Python code:
 - (a) to print all the keys.
 - (b) to print all the values.
 - (c) to print all the keys and values pairs.
 - (d) to print all the keys and values pairs in order of key.
 - (e) to print all the keys and values pairs in order of value.

5. Letter counts using dictionaries:
Remember to consider spaces as a special case.
 - (a) Write a function that takes as input a string and returns the most common letter in the string.
 - (b) Write a function that takes as input a string and returns a dictionary of letter counts.
 - (c) Write a function that takes as input a string and prints a histogram of letter counts.

6. Write a function that takes a person's first and last names as input and
 - (a) uses lists to return a list of the common letters in the first and last names (the intersection).
 - (b) uses sets to return a set that is the intersection of the characters in the first and last names.
 - (c) uses sets to return the set that is the symmetric difference between the first and last names.